CS 686 – Assignment 3 – Code is attached in the assignment

1. Decision Tree

Breakdown of Results

- Average Validation Accuracies:
    - Threshold 10: 0.7770
    - Threshold 20: 0.8010
    - Threshold 40: 0.8190 (The highest accuracy)
    - Threshold 80: 0.8140
    - Threshold 160: 0.8010
- Best Threshold: 40
- Final Test Accuracy: 0.8174

Discussion of Results

Based on the result, it was indicated that the sweet spot is at 40.

- Lower Thresholds (10, 20): These led to a reduced accuracy due to an over-grown tree. It is not only overfitting but also divided on very small data blocks (only 10 or 20 houses). The model simply was learning particular houses in the training set as opposed to general rules.
- Increased Thresholds (80, 160): This led to low accuracy once again due to premature termination of the tree. This is called underfitting. This model was not that complex to reflect the housing market.
- The Best Model: A model with a threshold of 40 was an effective one that removed the noise and retained the important patterns.

RESULT OF CODE

--- 10-Fold Cross Validation ---

Threshold 10: Avg Val Acc = 0.7770

Threshold 20: Avg Val Acc = 0.8010

Threshold 40: Avg Val Acc = 0.8190

Threshold 80: Avg Val Acc = 0.8140

Threshold 160: Avg Val Acc = 0.8010

Best Threshold: 40

Test Accuracy: 0.8174

Tree Structure:

```
Neighborhood = NAmes:
    BsmtExposure = NA:
        class = low
    BsmtExposure = Av:
        class = low
    BsmtExposure = No:
        FireplaceQu = Fa:
            class = low
        FireplaceQu = NA:
            LotConfig = Inside:
                GarageCond = Fa:
                    class = low
                GarageCond = NA:
                    class = low
                GarageCond = TA:
                    Exterior2nd = Other:
                        class = low
                    Exterior2nd = HdBoard:
                        class = low
                    Exterior2nd = ImStucc:
                        class = low
                    Exterior2nd = MetalSd:
                        class = low
                    Exterior2nd = Wd Sdng:
                        class = low
                    Exterior2nd = BrkFace:
```

class = low

Exterior2nd = CBlock:

class = low

Exterior2nd = Wd Shng:

class = low

Exterior2nd = CmentBd:

class = low

Exterior2nd = VinylSd:

class = low

Exterior2nd = AsbShng:

class = low

Exterior2nd = Stone:

class = low

Exterior2nd = Plywood:

class = low

Exterior2nd = Stucco:

class = low

Exterior2nd = Brk Cmn:

class = low

Exterior2nd = AsphShn:

class = low

GarageCond = Po:

class = low

GarageCond = Gd:

class = low

GarageCond = Ex:

class = high

LotConfig = Corner:

class = low

LotConfig = FR2:

class = high

LotConfig = CulDSac:

class = high

LotConfig = FR3:

class = low

FireplaceQu = TA:

class = low

FireplaceQu = Po:

class = low

FireplaceQu = Gd:

class = low

FireplaceQu = Ex:

class = high

BsmtExposure = Mn:

class = low

BsmtExposure = Gd:

class = high

Neighborhood = NridgHt:

BsmtExposure = NA:

class = high

BsmtExposure = Av:

class = high

BsmtExposure = No:

class = high

BsmtExposure = Mn:

class = high

BsmtExposure = Gd:

class = high

Neighborhood = MeadowV:

class = low

Neighborhood = SawyerW:

ExterQual = Fa:

class = low

ExterQual = Gd:

class = high

ExterQual = Ex:

class = high

ExterQual = TA:

class = low

Neighborhood = NWAmes:

Exterior1st = HdBoard:

class = high

Exterior1st = ImStucc:

class = high

Exterior1st = MetalSd:

class = high

Exterior1st = BrkComm:

class = high

Exterior1st = Wd Sdng:

class = high

Exterior1st = BrkFace:

class = low

Exterior1st = CBlock:

class = high

Exterior1st = CemntBd:

class = high

Exterior1st = VinylSd:

class = high

Exterior1st = AsbShng:

class = high

Exterior1st = Stone:

class = high

Exterior1st = Plywood:

class = high

Exterior1st = WdShing:

class = high

Exterior1st = Stucco:

class = high

Exterior1st = AsphShn:

class = high

Neighborhood = Crawfor:

class = high

Neighborhood = Edwards:

KitchenQual = Fa:

```
            class = low
        KitchenQual = Gd:
            class = high
        KitchenQual = Ex:
            class = high
        KitchenQual = TA:
            GarageFinish = RFn:
                class = low
            GarageFinish = NA:
                class = low
            GarageFinish = Unf:
                class = low
            GarageFinish = Fin:
                class = high
Neighborhood = NoRidge:
    class = high
Neighborhood = Timber:
    class = high
Neighborhood = Gilbert:
    BsmtFinType1 = NA:
        class = high
    BsmtFinType1 = GLQ:
        class = high
    BsmtFinType1 = ALQ:
        class = high
    BsmtFinType1 = BLQ:
```

```
            class = high
        BsmtFinType1 = Unf:
            class = high
        BsmtFinType1 = Rec:
            class = low
        BsmtFinType1 = LwQ:
            class = low
Neighborhood = IDOTRR:
    class = low
Neighborhood = NPkVill:
    class = low
Neighborhood = Blmngtn:
    class = high
Neighborhood = SWISU:
    class = low
Neighborhood = ClearCr:
    class = high
Neighborhood = Veenker:
    class = high
Neighborhood = OldTown:
    Exterior1st = HdBoard:
        class = low
    Exterior1st = ImStucc:
        class = low
    Exterior1st = MetalSd:
        class = low
```

Exterior1st = BrkComm:

class = low

Exterior1st = Wd Sdng:

class = low

Exterior1st = BrkFace:

class = low

Exterior1st = CBlock:

class = low

Exterior1st = CemntBd:

class = high

Exterior1st = VinylSd:

class = low

Exterior1st = AsbShng:

class = low

Exterior1st = Stone:

class = low

Exterior1st = Plywood:

class = low

Exterior1st = WdShing:

class = high

Exterior1st = Stucco:

class = low

Exterior1st = AsphShn:

class = low

Neighborhood = CollgCr:

KitchenQual = Fa:

```
        class = high
KitchenQual = Gd:
    MSZoning = RM:
        class = low
    MSZoning = C (all):
        class = high
    MSZoning = FV:
        class = high
    MSZoning = RH:
        class = high
    MSZoning = RL:
        GarageFinish = RFn:
            BsmtCond = Fa:
                class = high
            BsmtCond = NA:
                class = high
            BsmtCond = TA:
                LotShape = Reg:
                    class = high
                LotShape = IR1:
                    class = high
                LotShape = IR2:
                    class = high
                LotShape = IR3:
                    class = high
            BsmtCond = Po:
```

```
                    class = high
                BsmtCond = Gd:
                    class = high
            GarageFinish = NA:
                class = high
            GarageFinish = Unf:
                class = low
            GarageFinish = Fin:
                class = high
    KitchenQual = Ex:
        class = high
    KitchenQual = TA:
        class = low
Neighborhood = Mitchel:
    class = low
Neighborhood = BrkSide:
    Exterior1st = HdBoard:
        class = high
    Exterior1st = ImStucc:
        class = low
    Exterior1st = MetalSd:
        class = low
    Exterior1st = BrkComm:
        class = low
    Exterior1st = Wd Sdng:
        class = low
```

```
    Exterior1st = BrkFace:
        class = high
    Exterior1st = CBlock:
        class = low
    Exterior1st = CemntBd:
        class = low
    Exterior1st = VinylSd:
        class = low
    Exterior1st = AsbShng:
        class = low
    Exterior1st = Stone:
        class = low
    Exterior1st = Plywood:
        class = low
    Exterior1st = WdShing:
        class = low
    Exterior1st = Stucco:
        class = low
    Exterior1st = AsphShn:
        class = low
Neighborhood = StoneBr:
    class = high
Neighborhood = BrDale:
    class = low
Neighborhood = Somerst:
    Exterior2nd = Other:
```

class = high

Exterior2nd = HdBoard:

class = high

Exterior2nd = ImStucc:

class = low

Exterior2nd = MetalSd:

class = high

Exterior2nd = Wd Sdng:

class = high

Exterior2nd = BrkFace:

class = high

Exterior2nd = CBlock:

class = high

Exterior2nd = Wd Shng:

class = high

Exterior2nd = CmentBd:

class = high

Exterior2nd = VinylSd:

class = high

Exterior2nd = AsbShng:

class = high

Exterior2nd = Stone:

class = high

Exterior2nd = Plywood:

class = high

Exterior2nd = Stucco:

class = high

Exterior2nd = Brk Cmn:

class = high

Exterior2nd = AsphShn:

class = high

Neighborhood = Sawyer:

FireplaceQu = Fa:

class = high

FireplaceQu = NA:

class = low

FireplaceQu = TA:

class = low

FireplaceQu = Po:

class = low

FireplaceQu = Gd:

class = high

FireplaceQu = Ex:

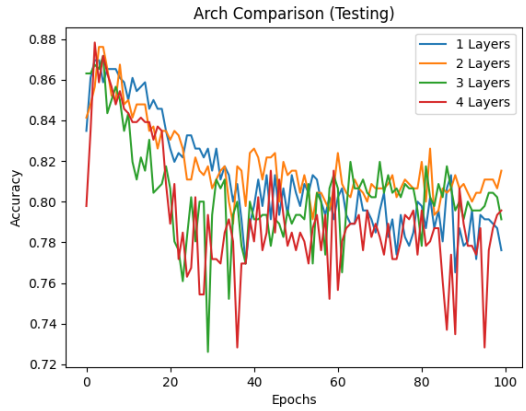
class = low

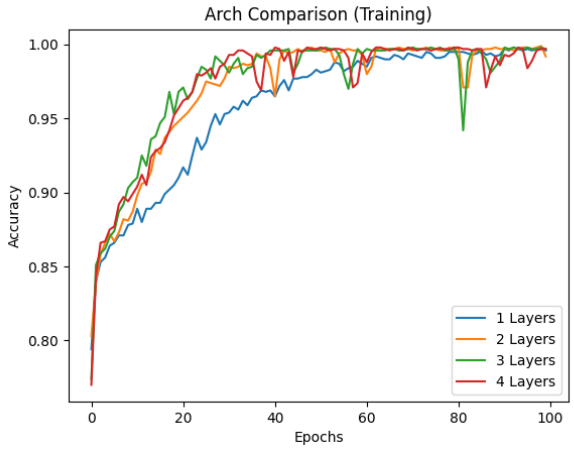
Neighborhood = Blueste:

class = low

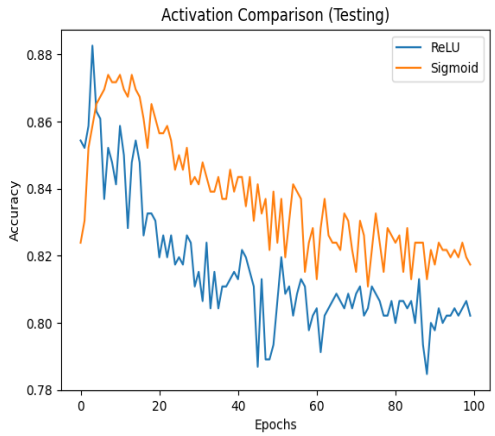
2. Neural Networks – In all the test graphs, there is a slight oscillation in the accuracy curves since the model was being trained on a mini-batch size of 10 which was defined in the skeleton code. This stochasticity is common to mini-batch gradient descent, but the general increasing direction is an assurance that the model is learning.

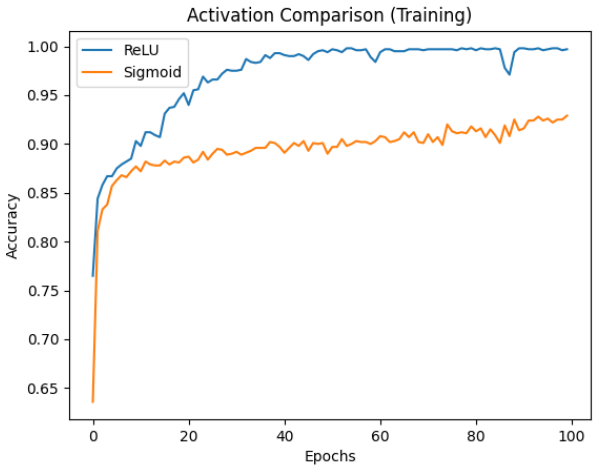
• Architecture comparison: The training accuracy increased progressively at higher rates as the number of hidden layers was increased to four. By incorporation of more layers, the network can learn more complex and nonlinear patterns in the real-estate data. The gap between training and testing accuracy was broader with the 4 layer model. This implies that unregularized deeper networks have higher chances of overfitting. The 4-layer training line will still be higher than the 1-layer line



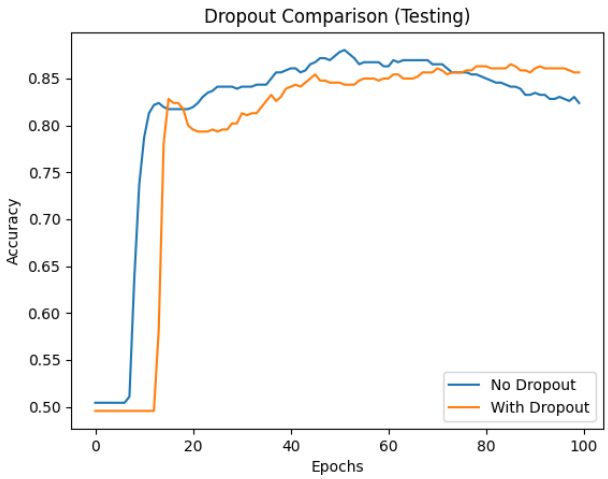


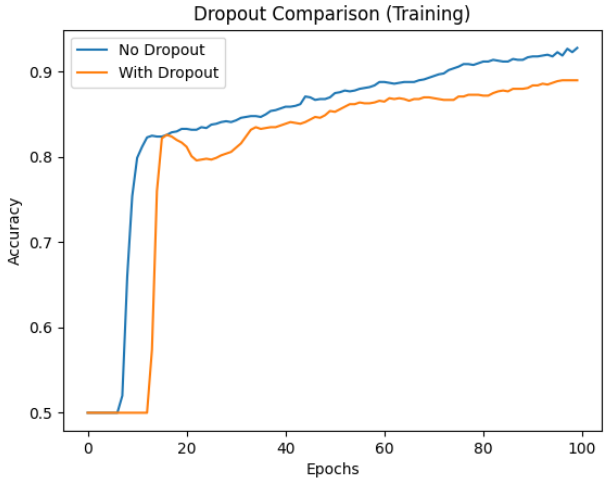
• Activation function comparison: ReLU has a steep curve, which attains high accuracy (80%+), fast while, Sigmoid has less steep or slower ascending curve that may not be able to crest 60-70 percent in the same number of epochs. ReLU activation function greatly performed better during the training and testing than Sigmoid function. ReLU eliminates the problem of vanishing gradient, which enables the network to learn effectively initially. Conversely, the gradients of the Sigmoid function are very low when the input or outputs are high or low, respectively, and thus convergence is much slower, and the final accuracy is lower with only 100 epochs. ReLU will still be higher than Sigmoid.

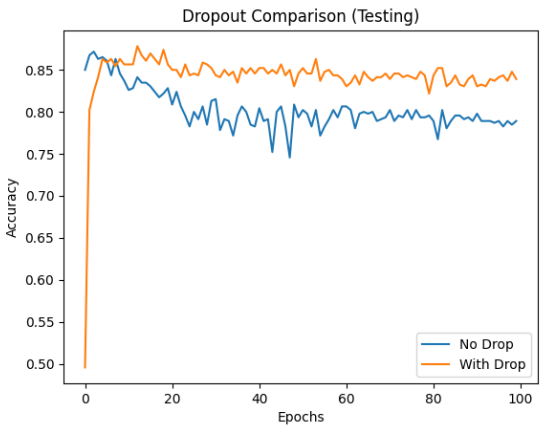


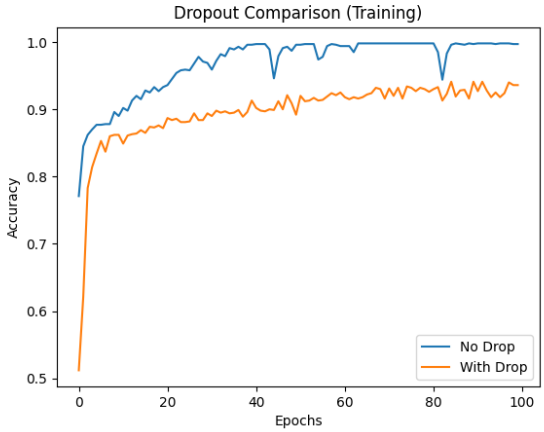


- • Dropout comparison: No Dropout has extremely large about 98 -100 percent training accuracy while, Dropout has the testing accuracy that should be smoother and closer to the training line. The training accuracy was better when dropout was not applied. The training task is further complicated by dropout randomly disabling nodes. Nonetheless, in the case of dropout, the testing accuracy should remain more or stable in comparison to the training line. This shows that dropout is a regularizer that does not allow the 4-layer network to over-learn the use of a particular set of neurons but instead enables the network to better generalize to new house data. Dropout will still show a smaller gap between Train and Test.






Dropout Comparison (Testing)
Accuracy
Epochs
No Drop
With Drop
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0
20
40
60
80
100


Dropout Comparison (Training)
Accuracy
Epochs
No Drop
With Drop
1.0
0.9
0.8
0.7
0.6
0.5
0
20
40
60
80
100